Test sets, statistics and results

COS30049 A2, group HCMC_COS30049.1_4 (generated by `tools/test_report.py`)

2026-10-29 22:10 UTC, commit `00b4966`

What is measured

Every writer answers each prompt; code then checks the answer (`explorer/assistant.py`, `lm/eval_explain.py`). *Guardrail*: numbers only from the facts, comparison words that match the data, the right verdict (flagged, not flagged, not scored), the limitation line on every flag, no disallowed wording. *Fully correct*: the guardrail passes and the answer names the template's reasons, strongest first, each number on its own feature's line (features whose $|z|$ shows the same in the prompt may come in either order). *Macro*: the mean over cases, so common easy cases cannot carry the rate. *Nothing out*: fully correct where no feature stands out, the one case in which a writer must rank features itself. All figures are percentages. The template is the reference, so it scores 100 by construction; it checks that each set is consistent.

Overview: fully correct, by test set

Test set	Prompts	Template	152M r1	152M r2
DS7 v1 test (frozen, round 1)	445	100.0	87.0	96.9
DS10 (teammate's snapshot)	291	100.0	83.2	93.1
Random: real	500	100.0	88.4	95.0
Random: synthetic	500	100.0	77.2	96.4
Random: hard	500	100.0	43.2	94.4
Random: balanced	883	100.0	72.0	94.9
Crawled: DS1 test split	174	100.0	89.7	94.3
Crawled: DS2 test split	421	100.0	82.4	96.7
Crawled: DS10	334	100.0	76.3	97.6
Close calls	300	100.0	80.0	93.0
Extreme values	300	100.0	74.7	95.3
Maximal shift	300	100.0	18.7	89.7

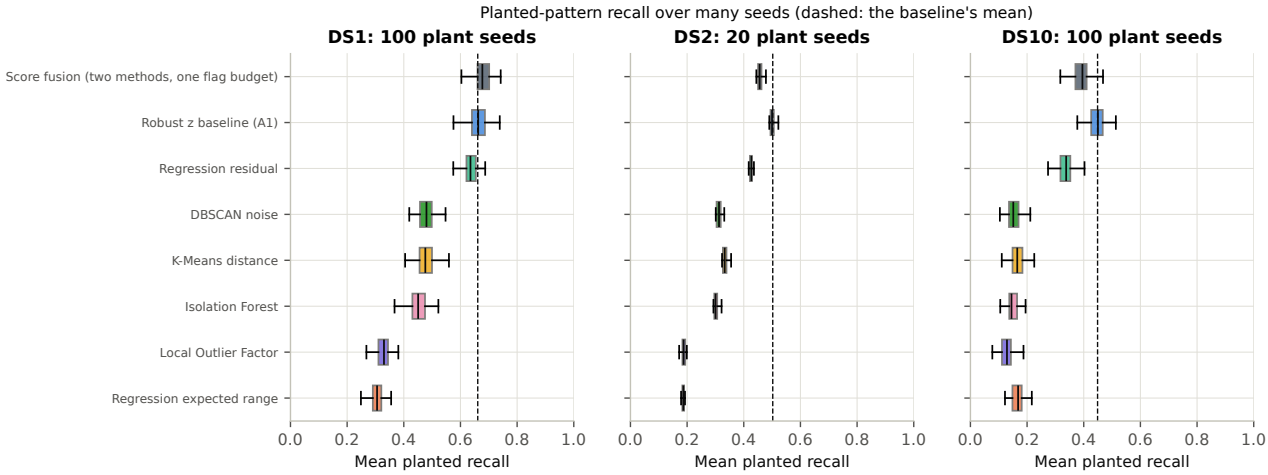
Statistics campaign

Distributions rather than single runs (requested 27 Oct). The scoring methods' planted patterns were replanted with many seeds (`explorer/distribution.py`); the classifiers were refitted with ten seeds and their test scores bootstrapped 1,000 times (`explorer/classify_distribution.py`). The language models were scored on 12 test sets with intervals and paired tests (extttlm/eval_stats.py).

Scoring methods: mean planted recall over plant seeds

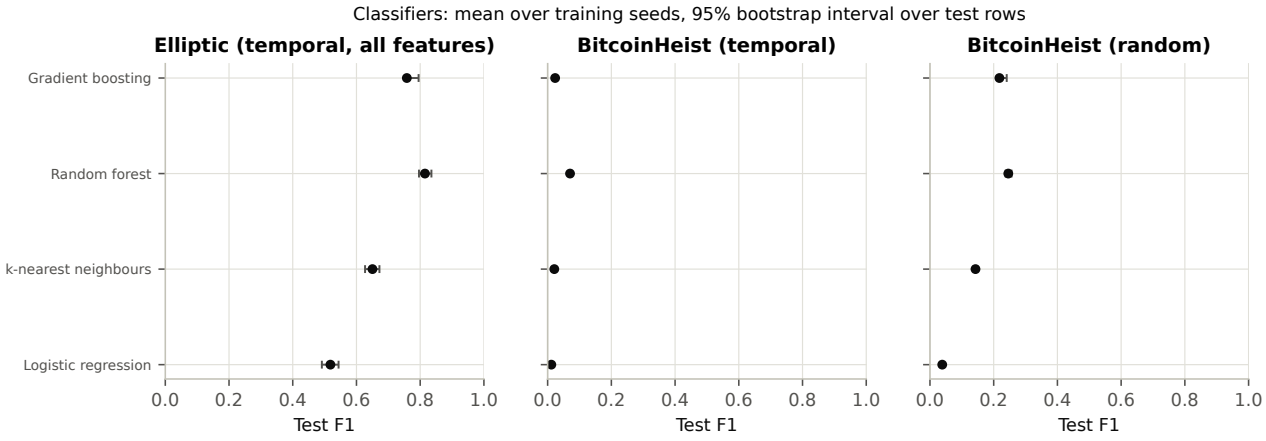
Data	Method	Seeds	Mean (sd)	95% of seeds	Beats baseline	Mean rank
DS1	Score fusion	100	0.679 (0.030)	0.617–0.726	77%	1.3
DS1	Robust-z baseline	100	0.661 (0.035)	0.588–0.720	0%	2.0
DS1	Regression residual	100	0.638 (0.025)	0.586–0.682	28%	2.7
DS1	DBSCAN noise	100	0.481 (0.032)	0.430–0.542	0%	4.6
DS1	K-Means distance	100	0.479 (0.033)	0.413–0.543	0%	4.8
DS1	Isolation Forest	100	0.452 (0.036)	0.386–0.522	0%	5.6

Data	Method	Seeds	Mean (sd)	95% of seeds	Beats baseline	Mean rank
DS1	Local Outlier Factor	100	0.329 (0.025)	0.286–0.376	0%	7.3
DS1	Regression range	100	0.306 (0.023)	0.261–0.347	0%	7.7
DS10	Robust-z baseline	100	0.449 (0.031)	0.381–0.507	0%	1.0
DS10	Score fusion	100	0.392 (0.030)	0.328–0.453	4%	2.0
DS10	Regression residual	100	0.336 (0.028)	0.278–0.392	0%	2.9
DS10	K-Means distance	100	0.169 (0.026)	0.129–0.222	0%	4.8
DS10	Regression range	100	0.167 (0.025)	0.125–0.210	0%	5.4
DS10	DBSCAN noise	100	0.154 (0.027)	0.114–0.206	0%	6.0
DS10	Isolation Forest	100	0.149 (0.023)	0.108–0.193	0%	6.3
DS10	Local Outlier Factor	100	0.128 (0.024)	0.087–0.170	0%	7.5
DS2	Robust-z baseline	20	0.502 (0.010)	0.491–0.524	0%	1.0
DS2	Score fusion	20	0.457 (0.008)	0.445–0.472	0%	2.0
DS2	Regression residual	20	0.425 (0.006)	0.413–0.434	0%	3.0
DS2	K-Means distance	20	0.334 (0.009)	0.324–0.353	0%	4.0
DS2	DBSCAN noise	20	0.314 (0.011)	0.301–0.337	0%	5.0
DS2	Isolation Forest	20	0.303 (0.009)	0.293–0.319	0%	6.0
DS2	Local Outlier Factor	20	0.188 (0.008)	0.173–0.198	0%	7.3
DS2	Regression range	20	0.186 (0.004)	0.179–0.192	0%	7.7



Classifiers: test F1 over training seeds, with a bootstrap of the test rows

Data and split	Model	Seeds	F1 mean (seed sd)	Bootstrap 95% CI, first seed	PR-A
Elliptic (temporal, all features)	Logistic regression	1	0.518 (0.000)	0.491–0.544	0
Elliptic (temporal, all features)	k-nearest neighbours	1	0.650 (0.000)	0.627–0.672	0
Elliptic (temporal, all features)	Random forest	10	0.815 (0.002)	0.796–0.835	0
Elliptic (temporal, all features)	Gradient boosting	10	0.758 (0.013)	0.755–0.795	0
BitcoinHeist (temporal)	Logistic regression	1	0.012 (0.000)	0.011–0.012	0
BitcoinHeist (temporal)	k-nearest neighbours	1	0.021 (0.000)	0.020–0.022	0
BitcoinHeist (temporal)	Random forest	10	0.070 (0.000)	0.068–0.075	0
BitcoinHeist (temporal)	Gradient boosting	10	0.024 (0.000)	0.023–0.024	0
BitcoinHeist (random)	Logistic regression	1	0.039 (0.000)	0.037–0.039	0
BitcoinHeist (random)	k-nearest neighbours	1	0.143 (0.000)	0.137–0.148	0
BitcoinHeist (random)	Random forest	10	0.246 (0.001)	0.238–0.256	0
BitcoinHeist (random)	Gradient boosting	10	0.218 (0.008)	0.224–0.241	0

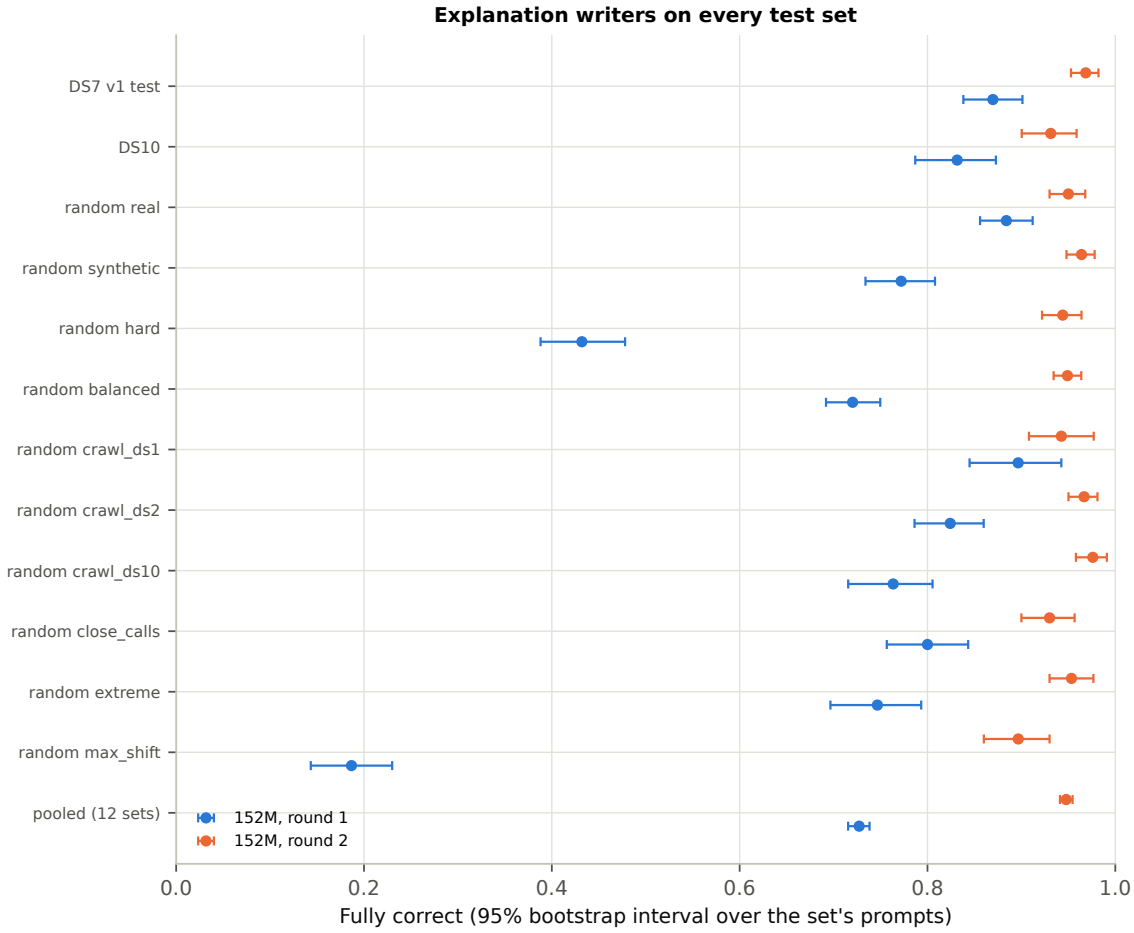


Explanation writers: fully correct with 95% bootstrap intervals, and paired tests

Test set	Writer	Rate	95% CI
pooled (12 sets)	152M, round 1	72.7	71.6–73.8
pooled (12 sets)	152M, round 2	94.8	94.1–95.4
glossary (key facts)	152M, round 1	22.5	13.5–31.5
glossary (key facts)	152M, round 2	48.3	37.0–59.1

152M round 1 minus round 2 on identical prompts (negative: round 2 better); McNemar's exact test counts the prompts only one of them got right.

Test set	Difference	95% CI	Only r1 right	Only r2 right	McNemar p
DS7 v1 test	-9.9	-13.0 to -6.7	9	53	<0.0001
DS10	-10.0	-13.8 to -5.8	5	34	<0.0001
random real	-6.6	-9.6 to -3.6	14	47	<0.0001
random synthetic	-19.2	-23.0 to -15.4	5	101	<0.0001
random hard	-51.2	-56.0 to -46.6	4	260	<0.0001
random balanced	-22.9	-25.8 to -19.9	8	210	<0.0001
random crawl_ds1	-4.6	-9.2 to -0.6	3	11	0.0574
random crawl_ds2	-14.3	-18.1 to -10.7	7	67	<0.0001
random crawl_ds10	-21.3	-26.0 to -17.1	3	74	<0.0001
random close_calls	-13.0	-17.7 to -8.7	8	47	<0.0001
random extreme	-20.7	-25.7 to -16.0	3	65	<0.0001
random max_shift	-71.0	-76.3 to -65.7	2	215	<0.0001
glossary (key facts)	-25.8	-37.5 to -13.3	–	–	–



The language models' test sets

DS7 v1 test (frozen, round 1)

Source. DS1 and DS2 transactions of the test-split addresses (DS7 splits by a seeded hash of the address).
How drawn. Round 1's DS7 build: up to 40 pairs per DS1 address and 10 per DS2 address, half flagged at most.
File. `data/processed/ds7/test.jsonl` (445 prompts)

Case	Prompts
unscored	39
flagged, 1 reason	64
flagged, 2 reasons	58
flagged, 3 reasons	28
flagged, 4 reasons	6
not flagged, nothing stands out	167
not flagged, 1 reason	62
not flagged, 2 reasons	16
not flagged, 3 reasons	5

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.3	100.0	97.8	87.6	87.0	72.7	86.8
152M, round 2	100.0	100.0	100.0	96.9	96.9	98.4	94.0

DS10 (teammate's snapshot)

Source. DS10: Le Quang Anh Hoang's independent Esplora snapshot (27 addresses), unseen by every model.
How drawn. The DS7 rule applied to DS10 by 'explorer.cli ds10-check'; frozen.

File. `data/processed/ds10/explain.jsonl` (291 prompts)

Case	Prompts
unscored	27
flagged, 1 reason	26
flagged, 2 reasons	54
flagged, 3 reasons	27
flagged, 4 reasons	2
not flagged, nothing stands out	102
not flagged, 1 reason	36
not flagged, 2 reasons	13
not flagged, 3 reasons	3
not flagged, 4 reasons	1

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.3	100.0	97.6	83.8	83.2	72.5	81.4
152M, round 2	100.0	100.0	99.7	93.1	93.1	91.2	90.2

Random: real

Source. DS7 v2 test split: DS1 and DS2 test-split addresses.
How drawn. 500 drawn at random (seed 50000), none in the frozen v1 test.
File. `data/processed/random_test/real.jsonl` (500 prompts)

Case	Prompts
unscored	33
flagged, 1 reason	47
flagged, 2 reasons	50
flagged, 3 reasons	23
flagged, 4 reasons	2
not flagged, nothing stands out	256
not flagged, 1 reason	66
not flagged, 2 reasons	20
not flagged, 3 reasons	3

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.2	100.0	97.6	89.2	88.4	85.4	84.0
152M, round 2	99.8	100.0	100.0	95.2	95.0	98.0	91.0

Build log (`reports/logs/build-random-test.log`, last run):

```
2026-10-26 21:32:39 INFO Step build-random-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-random-test
2026-10-26 21:32:40 INFO Randomized test set: 500 real prompts (from 1401 eligible), 500 synthetic and 500 hard: {'real': {'flagged': 122, 'not_fl
2026-10-26 21:32:40 INFO Step build-random-test finished in 1 s (exit 0)
```

Random: synthetic

Source. Made up: values, usual values and z drawn from settings randomTest, capped to realistic limits.
How drawn. 500 transactions (seed 50000); z from a mixture, values derived from z.
File. `data/processed/random_test/synthetic.jsonl` (500 prompts)

Case	Prompts
unscored	34
flagged, 1 reason	36
flagged, 2 reasons	70
flagged, 3 reasons	54
flagged, 4 reasons	27
not flagged, nothing stands out	79
not flagged, 1 reason	116
not flagged, 2 reasons	62
not flagged, 3 reasons	16
not flagged, 4 reasons	6

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	96.8	100.0	96.6	78.2	77.2	66.2	77.2
152M, round 2	98.6	100.0	99.8	97.6	96.4	93.5	97.5

Build log (`reports/logs/build-random-test.log`, last run):

```
2026-10-26 21:32:39 INFO Step build-random-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-random-test
2026-10-26 21:32:40 INFO Randomized test set: 500 real prompts (from 1401 eligible), 500 synthetic and 500 hard: {'real': {'flagged': 122, 'not_fl
2026-10-26 21:32:40 INFO Step build-random-test finished in 1 s (exit 0)
```

Random: hard

Source. Made up, explained for any of the nine methods.
How drawn. 500 (seed 50001): 289 non-baseline, about half reworded and half shuffled.
File. `data/processed/random_test/hard.jsonl` (500 prompts)

Case	Prompts
unscored	44
flagged, nothing stands out	13
flagged, 1 reason	46
flagged, 2 reasons	60
flagged, 3 reasons	38
flagged, 4 reasons	14
not flagged, nothing stands out	73
not flagged, 1 reason	105
not flagged, 2 reasons	69
not flagged, 3 reasons	29
not flagged, 4 reasons	9

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	85.2	99.6	57.2	46.0	43.2	33.7	44.2
152M, round 2	99.4	100.0	99.0	95.0	94.4	90.2	83.7

Build log (`reports/logs/build-random-test.log`, last run):

```
2026-10-26 21:32:39 INFO Step build-random-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-random-test
2026-10-26 21:32:40 INFO Randomized test set: 500 real prompts (from 1401 eligible), 500 synthetic and 500 hard: {'real': {'flagged': 122, 'not_fl
2026-10-26 21:32:40 INFO Step build-random-test finished in 1 s (exit 0)
```

Random: balanced

Source. Every transaction of DS10 and of DS2's 173 test-split addresses (46,513 explained).
How drawn. De-duplicated, other test sets removed, up to 100 per case (seed 50002).
File. `data/processed/random_test/balanced.jsonl` (883 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	100
flagged, 2 reasons	100
flagged, 3 reasons	100
flagged, 4 reasons	95
not flagged, nothing stands out	100
not flagged, 1 reason	100
not flagged, 2 reasons	100
not flagged, 3 reasons	100
not flagged, 4 reasons	87

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.3	100.0	93.1	72.4	72.0	73.9	82.0
152M, round 2	100.0	100.0	100.0	94.9	94.9	95.2	97.0

Build log (`reports/logs/build-balanced-test.log`, last run):

```
2026-10-26 21:43:37 INFO Step build-balanced-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-balanced-test
2026-10-26 21:43:48 INFO DS2 test split: 173 addresses, 40728 transactions explained
2026-10-26 21:43:50 INFO DS10: 27 addresses, 5785 transactions explained
2026-10-26 21:43:51 INFO Balanced test set: 883 prompts; per case (unique in pool, drawn): {'flagged, 1 reason': (1426, 100), 'flagged, 2 reasons'
2026-10-26 21:43:51 INFO Step build-balanced-test finished in 13 s (exit 0)
```

Crawled: DS1 test split

Source. DS1 (our Esplora snapshot): the test-split addresses' transactions.
How drawn. Up to 60 per case (seed 50003); no prompt of an earlier set.
File. `data/processed/random_test/crawl_ds1.jsonl` (174 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	11
flagged, 2 reasons	7
flagged, 3 reasons	4
flagged, 4 reasons	2
not flagged, nothing stands out	60
not flagged, 1 reason	60
not flagged, 2 reasons	26
not flagged, 3 reasons	3

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.4	100.0	98.9	90.2	89.7	80.6	86.7
152M, round 2	100.0	100.0	100.0	94.3	94.3	94.6	85.0

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons'
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons'
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Crawled: DS2 test split

Source. DS2 (our Esplora crawl of 4,012 addresses): the 173 test-split addresses.
How drawn. Up to 60 per case (seed 50003); no prompt of an earlier set.
File. `data/processed/random_test/crawl_ds2.jsonl` (421 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	60
flagged, 2 reasons	60
flagged, 3 reasons	60
not flagged, nothing stands out	60
not flagged, 1 reason	60
not flagged, 2 reasons	60
not flagged, 3 reasons	60

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.8	100.0	99.3	82.9	82.4	84.6	86.7
152M, round 2	100.0	100.0	99.8	96.7	96.7	97.1	91.7

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Crawled: DS10

Source. DS10 (the teammate's crawl): all 27 addresses.
How drawn. Up to 60 per case (seed 50003); no prompt of an earlier set.
File. `data/processed/random_test/crawl_ds10.jsonl` (334 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	30
flagged, 2 reasons	48
flagged, 3 reasons	15
not flagged, nothing stands out	60
not flagged, 1 reason	60
not flagged, 2 reasons	60
not flagged, 3 reasons	60

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	100.0	100.0	97.3	76.3	76.3	80.0	78.3
152M, round 2	100.0	100.0	99.7	97.6	97.6	98.3	91.7

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Close calls

Source. DS1, DS2 and DS10 test-split transactions.
How drawn. 300 whose listed |z| differ by less than 0.3, so order and pair are hardest.
File. `data/processed/random_test/close_calls.jsonl` (300 prompts)

Case	Prompts
flagged, 3 reasons	2
not flagged, nothing stands out	276
not flagged, 2 reasons	17
not flagged, 3 reasons	5

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	97.7	100.0	94.3	81.7	80.0	54.2	82.2
152M, round 2	100.0	100.0	98.7	93.0	93.0	98.1	92.4

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Extreme values

Source. Made up, one extreme regime per transaction.
How drawn. 300: 1-100 sat or about 1,000 BTC amounts, minute or multi-year gaps, 80-200 transactions a week.
File. `data/processed/random_test/extreme.jsonl` (300 prompts)

Case	Prompts
unscored	23
flagged, 1 reason	16
flagged, 2 reasons	44
flagged, 3 reasons	33
flagged, 4 reasons	5
not flagged, nothing stands out	64
not flagged, 1 reason	61
not flagged, 2 reasons	45
not flagged, 3 reasons	7
not flagged, 4 reasons	2

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	89.0	100.0	99.0	83.3	74.7	68.1	70.3
152M, round 2	98.3	100.0	99.7	97.0	95.3	96.0	87.5

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Maximal shift

Source. Made up, for methods other than the baseline.
How drawn. 300: every prompt reworded, its facts shuffled, the flag independent of the features.
File. `data/processed/random_test/max_shift.jsonl` (300 prompts)

Case	Prompts
unscored	28
flagged, nothing stands out	18
flagged, 1 reason	24
flagged, 2 reasons	21
flagged, 3 reasons	17
flagged, 4 reasons	4
not flagged, nothing stands out	40
not flagged, 1 reason	50
not flagged, 2 reasons	51
not flagged, 3 reasons	37
not flagged, 4 reasons	10

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	72.3	96.7	27.3	20.0	18.7	14.6	8.6
152M, round 2	99.3	100.0	94.3	90.3	89.7	89.3	62.1

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Glossary test (open questions)

Source. `tools/make_glossary.py`: 74 held-out phrasings of the 65 taught topics, each with the key facts a correct answer must state.
How drawn. One or two per topic, written differently from the taught questions; round 1's five test questions are among them.

Writer	Key facts stated	Fully right	ROUGE-L
152M, round 1	22.5	20.3	0.287
152M, round 2	48.3	43.2	0.480